

Nick Ioannidis

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EDUCATION

Simon Fraser University
Ph.D. in Computer Science

Jan. 2026 – Present
Supervisor: Prof. Xue Bin (Jason) Peng

University of British Columbia
M.Sc. in Computer Science

Sep. 2023 – May 2025
Supervisor: Prof. Michiel van de Panne

Thesis: *Viability Estimation for Diffusion-Based Planning*

University of British Columbia

B.A.Sc. in Engineering Physics, with Distinction

Sep. 2017 – Apr. 2023

PUBLICATIONS

- [1] **N. Ioannidis**, Daniele Reda, Setareh Cohan, and Michiel van de Panne. Diffusion-based planning with learned viability filters. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 8(4):1–23, 2025
- [2] Tiffany Matth , **N. Ioannidis**, and Michiel van de Panne. Walk this way: Imitation-free reinforcement learning of flexibly-constrained walking controllers. *Proceedings of the ACM on Computer Graphics and Interactive Techniques*, 8(4):1–19, 2025
- [3] **N. Ioannidis**, Jonathan Wilder Lavington, and Mark Schmidt. An empirical study of non-uniform sampling in off-policy reinforcement learning for continuous control. In *Deep RL Workshop, NeurIPS 2021*, 2021

RESEARCH EXPERIENCE

Motion Retargeting for Humanoid Robots
SFU · GrUVi Lab

May 2025 – Dec. 2025
Supervisor: Prof. Xue Bin (Jason) Peng

- Developed a motion retargeting method that produces higher-quality reference motions for humanoid robot control policies. [\[Under Review\]](#)

Humanoid Teleoperation via Three-Point Tracking
Robotics and AI Institute (RAI)

May 2025 – Dec. 2025

- Developing a learning-based system for whole-body humanoid teleoperation from three-point tracking. [\[Ongoing Project\]](#)

Planning and Control for Humanoid Characters

UBC · Motion Control and Character Animation Group (MOCCA)

May 2024 – Apr. 2025
Supervisor: Prof. Michiel van de Panne

- Used diffusion models for footstep planning of humanoid characters, with learned viability filters to ensure generated plans satisfy constraints and remain feasible. [\[PACMCGIT 2025\]](#)
- Learned flexibly-constrained walking controllers for humanoid characters with reinforcement learning, without relying on motion-capture imitation. [\[PACMCGIT 2025\]](#)

Non-Uniform Sampling for Off-Policy Reinforcement Learning

UBC · Department of Computer Science

May 2021 – Aug. 2021; May 2022 – Aug. 2022
Supervisor: Prof. Mark Schmidt

- Conducted an empirical study of non-uniform replay sampling strategies for off-policy RL in continuous control, training agents on MuJoCo tasks from the DeepMind Control Suite on HPC clusters. [\[NeurIPS 2021 Deep RL Workshop\]](#)

INDUSTRY EXPERIENCE

Machine Learning Engineer
UBC MRI Research Centre

Jun. 2020 – Apr. 2021
Vancouver, Canada

- Implemented and trained 3D volumetric segmentation architectures (3D U-Net, V-Net) in PyTorch.
- Studied the physical properties of microstructures to generate realistic synthetic samples, and used them to augment a sparse scan dataset (MATLAB, Julia).

Software Engineer (Co-op)

Craft Metrics

Jan. 2019 – Apr. 2019
Vancouver, Canada

- Implemented a backend service for customer provisioning in Go.
- Extended a real-time data pre-processing system in Python and integrated it with the main data pipeline.

R&D Electrical Engineer

Recycling Alternative

• Reconfigured compost reactors and built a Python data-collection system: a Raspberry Pi coordinating several Arduinos.

• Designed per-reactor control loops monitoring temperature, humidity, CO₂ and ammonia under varying initial conditions.

Jun. 2018 – Sep. 2018

Vancouver, Canada

SELECTED PROJECTS

Grounding LLM Plans via Simulation Context

UBC · CPSC 532V: Commonsense Reasoning in NLP

• Improved LLM planning by feeding back the results of executing generated plans in a simulated chess environment.

Jan. 2024 – May 2024

Robot-to-Robot Transfer for Quadruped Locomotion

UBC · CPSC 554X: Machine Learning and Signals

• Trained RL imitation policies on Open X-Embodiment motions to transfer locomotion across quadruped morphologies.

Sep. 2023 – Dec. 2023

Automatic Curriculum Generation for Hard-Exploration Tasks in Minecraft

UBC · CPSC 532S: Multimodal Learning with Vision, Language and Sound

• Used GPT-3 to generate task curricula for a Minecraft agent and implemented PPO with Self-Imitation Learning.

Sep. 2022 – Dec. 2022

Alzheimer’s Disease Classification with Transformer-Based Language Models

UBC · Canary Cognition Research Group

• Applied transfer learning and fine-tuning to transformer language models, improving over baseline performance on Alzheimer’s disease classification from a small target dataset.

Sep. 2021 – Apr. 2022

Supervisor: Prof. Hyeju Jang

Open Sim2Real: A Low-Cost Robotic Platform for RL Research

UBC · ENPH 479: Engineering Physics Capstone

• Built a monopod robot based on the Open Dynamic Robot Initiative design and trained PPO and SAC policies that achieved standing and balancing both in simulation and on the physical robot.

Sep. 2021 – Apr. 2022

Software Lead, UBC Open Robotics

UBC Engineering Design Team

• Led ROS/Python control software for a gripper-equipped TurtleBot performing household tasks, placing 2nd in the RoboCup@Home Education international competition.

Sep. 2019 – May 2021

TEACHING

Graduate TA, CPSC 533V: Learning to Move (UBC)

Graduate TA, CPSC 340: Machine Learning and Data Mining (UBC)

Graduate TA, CPSC 314: Computer Graphics (UBC)

Undergraduate TA, ENPH 353: Engineering Physics Project I (UBC)

2024

2024

2023

2023

Sep. 2024 – Dec. 2024

Jan. 2024 – Apr. 2024

Sep. 2023 – Dec. 2023

Sep. 2022 – Apr. 2023

SCHOLARSHIPS AND AWARDS

NSERC Undergraduate Student Research Award (\$6,000)

NSERC Experience Award (\$4,500)

UBC Dean’s Honour List

2022

2019

2019

SKILLS

Programming

Robotics & ML

Hardware

Languages

Python, C++ , MATLAB, Julia, Go

PyTorch, Isaac Lab, MuJoCo, PyBullet, ROS

NVIDIA Jetson, Raspberry Pi, Arduino, STM32, CAD (Onshape, SolidWorks)

English, Greek